

Experience

Applied Physics Laboratory (JHU)

Laurel, MD

Combat System Analyst & Mechanical Engineer – Air & Missile Defense Sector

August 2023 – Present

- Redesigning and prototyping portions of reentry bodies for greater impact precision.
- Assisted in rapid integration and testing of RF equipment into unmanned surface vessels.
- Streamlined data analysis and reporting of Aegis & TSCE ship testing & events. Reported on weapon control system data during tests and real-world engagements. Helped lead analysis efforts for test events.
- Modified LPBF machine for greater thermal management & control pertinent to printing shape memory alloys.
- Led research project in additive manufacturing of graded metal foams.

Mechanosynthesis Group (MIT)

Cambridge, MA

Graduate Research Assistant

September 2021 – June 2023

- Finalized and characterized a modular LPBF system to enable in house metal printing.
- Created a novel LPBF-inkjet system to enable printing compositionally graded parts with refractory alloys.

Pure Watercraft

Seattle, WA

Mechanical Engineer

February 2021 – July 2021

- Created a rig and performed log strike testing on outboard motors to ensure user safety.
- Developed equipment for system components & PCBs. Designed and built a dynamometer to test outboards.

Impact and Crashworthiness Lab (MIT)

(Remote)

Undergraduate Researcher

May 2020 – December 2020

- Developed physics-informed neural networks to solve unsteady heat transfer problems and supplant finite element solvers in modelling thermal runaway of damaged lithium-ion batteries.

Pure Watercraft

Seattle, WA

Mechatronics Intern

June 2020 – August 2020

- Developed test equipment to validate the BMS, battery control boards, and throttle construction.
- Designed and implemented data acquisition unit to analyze gearbox pressures and inform decisions on its valving.

BD Medical - Advanced Diabetes Care

Andover, MA

Research and Development Intern

June 2019 - August 2019

- Prototyped automated testing equipment to improve manufacturing efficiency of latest insulin delivery device.
- Supported the development of a new leak rate test method for critical modules of new insulin delivery device.

Musashi Auto Parts - Michigan

Battle Creek, MI

Engineering Intern

June 2017 - August 2017 & June 2018 - August 2018

- Designed and fabricated a computer-controlled tool cart to aid in machine installation.
- Determined the acceptable leak rate of gear assemblies and generated production specifications to reduce bad parts.

Education

Johns Hopkins University

Baltimore, MD

- Coursework in materials science & solid mechanics.

September 2024 - June 2026

Massachusetts Institute of Technology

Cambridge, MA

B.S. / M.S. in Mechanical Engineering

September 2017 - June 2023

- Exchange student at ETH Zürich, Switzerland in 2020.

Other Projects

Maryland Zoo Enrichment Engineering

April 2025 – Present

- Helping engineer enrichment opportunities for animals & creating sensor packages to evaluate efficacy.
- Mentoring undergraduates on embedded system development.

MIT Solar Electric Vehicle Team

September 2018 – December 2020

- Designed & manufactured parts of the mechanical system. Trained teammates on machining, welding, and FEA.
- Spearheaded team sponsorship efforts leading to an acquisition of over \$90k to support team operations.

Skills

- | | | |
|--------------|----------|-----------------|
| • SolidWorks | • Comsol | • Machining |
| • Matlab | • Creo | • Welding |
| • Python | • Abaqus | • Arduino/ESP32 |

Publications

- (Under review) Kutschke, Z., Wolfe, K., Birol, E., Kutzer, M., Mueller, J. (2026) *Rapid 3D Printing of Thin-Walled Structures via Line Extrusion*. International Journal of Extreme Manufacturing.
- Kutschke, Z., Li, Z., Berlia, R., Hemker, K., Weihs, T., & Mueller, J. (2025). *Metal extrusion additive manufacturing: A unified framework and comprehensive review of an emerging technology*. Journal of Manufacturing Processes. <https://doi.org/10.1016/j.jmapro.2025.10.078>
- Penny, R. W., Kutschke, Z., & Hart, A. J. (2025). *High-fidelity optical monitoring of laser powder bed fusion via aperture division multiplexing*. Npj Advanced Manufacturing. <https://doi.org/10.1038/s44334-025-00039-8>
- Tekoglu, E., Liao, S., Kutschke, Z., O'Brien, A. D., Lettiere, B., Li, J., & Hart, A. J. (2025). *Rapid Exploration of Nanoparticle-Modified Alloys in Metal Additive Manufacturing by Combining Inkjet Printing and Laser Powder Bed Fusion*. Additive Manufacturing Letters. <https://doi.org/10.1016/j.addlet.2025.100315>